

D-Lab: Water, Sanitation, and Hygiene

Final Project Report

Project

AC Hacking and Mapping

Team

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Mentors

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Abstract

This project is being completed in coordination with Cynthia Acquaye and her team of mentors at Delft University of Technology in the Netherlands. The project's goal is to explore the potential of using AC units to create clean drinking water. Through mapping the feasibility of this solution in Accra and investigating the production of condensate in a window AC unit, this project hopes to demonstrate the potential of this solution. The project maps factors such as temperature, humidity, and precipitation, as well as investigates electricity usage and maps water access to determine the areas best suited to deploy the project. This project also provides insights into modifying AC units to strike a balance between cooling capacity and drinking water production.

Completed Goals and Objectives

1. Developed mapping includes:
 - a. Climate Maps showing aggregate temperature and humidity
 - b. National Water Vulnerability Index
 - c. National Solution Feasibility Index
 - d. AC Donation Priority Map
2. AC Energy Balance
 - a. Developed a monte carlo model that can predict the water condensate produced per AC unit depending on the operational environment.
 - b. Demonstrate a performance balance for thermal management and water condensate production. Input variables include volumetric flow rate of air, inlet temperature, inlet relative humidity, outlet temperature, outlet humidity, coil temperature. Calculated outputs include estimated overall heat transfer coefficient, sensible heat factor values, two mass flow rates of water out of the ac systems using a (1) black box model and (2) bypass factor model.

3. Physical AC Unit
 - a. Designed a sensor-based system to track and analyze condensate produced.
 - b. Utilized a simple water condensate collection system via volumetric collection.
 - c. Hypothesize preliminary water treatment systems to add to the collection system.

Mapping of Water Vulnerability and AC Unit Potential in Ghana

Mapping Methods

Section Purpose: to help any future student on this project working with GIS methods, below is a **detailed** report of the methods I used for each final map! Feel free to skim, or reference in detail for the recreation of the maps.

1. Climate Maps
 - a. These maps utilized Copernicus ERA5 Climate data, from the years 2021-2025, each month. I imported the NetCDF for total precipitation first, creating a multidimensional raster layer, and then aggregating the multidimensional raster to get the average total precipitation across the data's time period. I repeated these steps for 2m dewpoint temperature and 2m temperature [figure 1]. I resampled each climate layer to a 1km cell size and clipped the raster to the Ghana Admin lines. To calculate relative humidity, I used the August-Roche-Magnus Formula. I used Claude AI to generate python code to convert kelvins to celsius and get the humidity [figure 2] using this formula. I then clipped this raster to the Ghana Admin lines.
2. Water Vulnerability
 - a. I applied for DHS data through the DHS Program website, and downloaded the GHGE8AFL.zip shapefile (I was granted geographic data access) and the GHHR8.CSV zipfile, which contained data from nearly 18,000 household surveys in 2022. The data comes in a form accessible only using statistical software, so my first task was to convert it into a format readable by excel and ArcGIS. I did not want to do this with the entire dataset, as it had over 400 variables included. Instead, I perused the data guide and located variables that I thought might be useful:
 - i. HV001 Cluster ID
 - ii. HV201 Main drinking water source
 - iii. HV202 Main water source for household use
 - iv. HV201A Water unavailable 1+ full day in the past 2 weeks
 - v. HV204 Time to water source in minutes
 - vi. HV206 Electricity
 - vii. HV207 Radio
 - viii. HV208 Television
 - ix. HV209 Refrigerator

- x. HV235 Location of water source
 - xi. HV270 Wealth index (categorical)
 - xii. HV271 Wealth index factor score
 - xiii. HV270A Wealth index urban/rural
 - xiv. HV271A Wealth index factor score urban/rural
 - b. Using Claude AI once more, I used R Studio (for the first time!) to read and sort the data. I manually checked that all my variables were in the spreadsheet, and then exported the csv. Next, I created a water vulnerability index based on the WHO definition of safely managed drinking water (1. Located on premises, 2. Available when needed, and 3. Free from contamination). I created 3 columns:
 - i. Unsafe Source as defined by variables HV201 32 (unprotected well), 42 (unprotected spring), 43 (river/dam/lake/pond/etc.), 51 (rainwater)
 - ii. No Water on Premises as defined by HV235 3 (elsewhere), HV204 998 (not on premises)
 - iii. Water Vulnerable as defined by having one or both vulnerabilities
 - c. Once each household had their vulnerability binary assigned to them (yes/no), I aggregated the data for each cluster, turning the table from ~18,000 rows into 618 cluster centroids from the geographic DHS data. Each row had the mean vulnerability index for each column, along with the mean electricity (yes/no binary → percentage of cluster households that have electricity) and mean wealth index (in number form from negative score to positive score). This analysis was all performed in R Studio, and the code can be found on [github](#). The new table (cluster_summary.csv) was imported into ArcGIS and joined with the cluster centroid layer. Null values were assumed to be 0 in the aggregation so as to not overestimate the severity of the water crisis. I then performed Inverse Distance Weighted (IDW) Interpolation on all 4 of my variables (percent electricity, no water on premises, mean wealth, and water vulnerability (eventually adding a 5th manually for unsafe water source)). I did this using a model for IDW, which performed the interpolation on each cluster layer based on a different column, created 4 different layers, and extracted them by mask from the Ghana Admin lines.
 - d. The vulnerability index itself shows which areas of Ghana are most water-vulnerable according to these WHO standards. The index includes:
 - i. meanWealth_idw (DHS 2022)
 - ii. noPremis_idw (DHS 2022)
 - iii. unsafeSource_idw (DHS 2022)
 - e. I performed Fuzzy Membership on all 3 layers. Large for noPremis and unsafeSource, Linear for meanWealth (wealth index values also contain negative values, trial and error revealed that this was throwing off the fuzzification, so I did linear with a minimum value of 168810.265265 (the largest value) and a maximum value of -188373.921875 (the smallest value) in order to simulate a "small" membership type). I did a resample to 0.01x0.01 cell size, and then ran a fuzzy overlay with all 3 layers (using overlay type "and").
3. National Feasibility

- a. The goal of this map was to create an index that shows where in Ghana this solution might be feasibly implemented. It used relative humidity, aggregated temperature, VIIRS nighttime lights, and percent electricity (IDW Interpolated from DHS data) to calculate this. The logic behind this map is that more humidity is good for AC unit condensate production, hot temperatures are where people would benefit from cooling technologies, and that VIIRS nighttime lights and electricity data complement each other as electricity proxies, and where people do not have electricity, the solution would not be able to be implemented, because electricity is needed for AC units to run. I ran fuzzy membership on all 4 input layers using the automatic settings. I then built a model to resample all of these layers and input them into a fuzzy overlay that resulted in the National Feasibility Index map.
4. AC Donation Priority
 - a. The goal of this map is to show areas that overlap in high electricity and vulnerability, showing the best areas to donate this solution to vulnerable households. First, I reclassified both the national vulnerability index from the second fuzzy overlay I performed (using the "or" function to show a wider range of areas that are vulnerable in any of the categories, not just all the categories. This cast a wider net on solution donation priority). I used equal interval classification. I did the same for the percent electricity layer. I used the CON tool and input the conditional raster: vulnerability_reclass_ghana where the value was equal to 3, or where the vulnerability was the highest level. I classified it based on electricity level. I input the true raster as percentElectricity_reclass, inputting the false or constant value as 0 so that I could symbolize it as grey and not empty on the map. The outcome was the AC Donation Priority map.

Mapping Limitations

Data availability: Ghana's limited spatial data infrastructure presented significant challenges. The Demographic and Health Surveys (DHS 2022) served as the primary household-level data source, but these cluster centroids are geo-displaced up to 2km in urban areas and 5km in rural areas, reducing locational precision. With the cancellation of USAID funding, future DHS surveys are uncertain, making this 2022 dataset potentially the last available for some time.

Methodological constraints: Interpolation of sparse cluster data creates an illusion of precision that does not exist in the underlying dataset. This analysis is most defensible at the national scale; neighborhood-level conclusions should be interpreted cautiously. Climate variables were averaged annually for simplicity, though seasonal variation between dry and rainy seasons would meaningfully affect condensate production estimates. Relative humidity was derived using the August-Roche-Magnus formula rather than sourced directly.

Proxy variables: Electricity access and wealth were approximated using proxy datasets (VIIRS nighttime lights and DHS wealth index) rather than household-level grid data, as the latter was unavailable.

Index completeness: While grounded in WHO standards, the vulnerability and feasibility indexes do not capture all relevant variables. Distance to water source and seasonal climate variation represent meaningful refinements for future work.

Ground-truthing: This analysis was conducted entirely remotely, with no ability to ground-truth the results. The cartographer's limited knowledge of the Ghanaian context, along with limited data availability, contributed to a potential gap between analysis and reality that should be addressed in further research. This further research should emphasize continued collaboration with actors in Ghana in order to maximize accuracy and avoid the simplification that accompanies the modeling and estimation used in this mapping process.

Final Maps

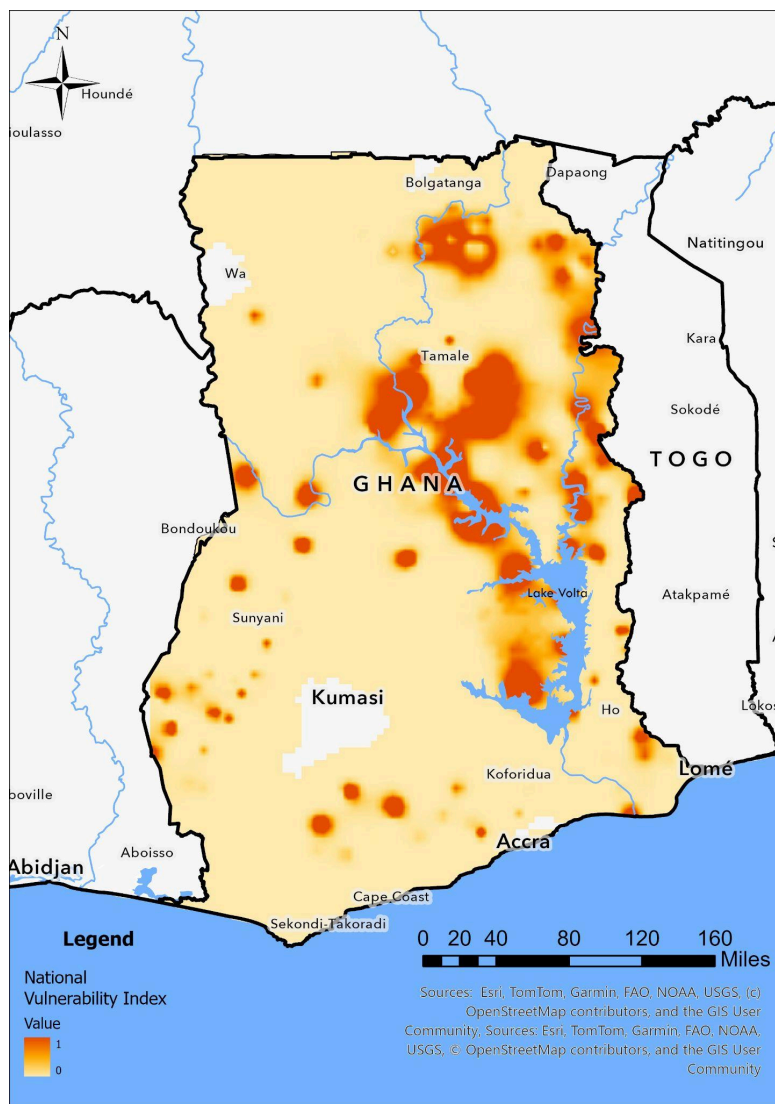


Figure: Ghana Water Vulnerability Index

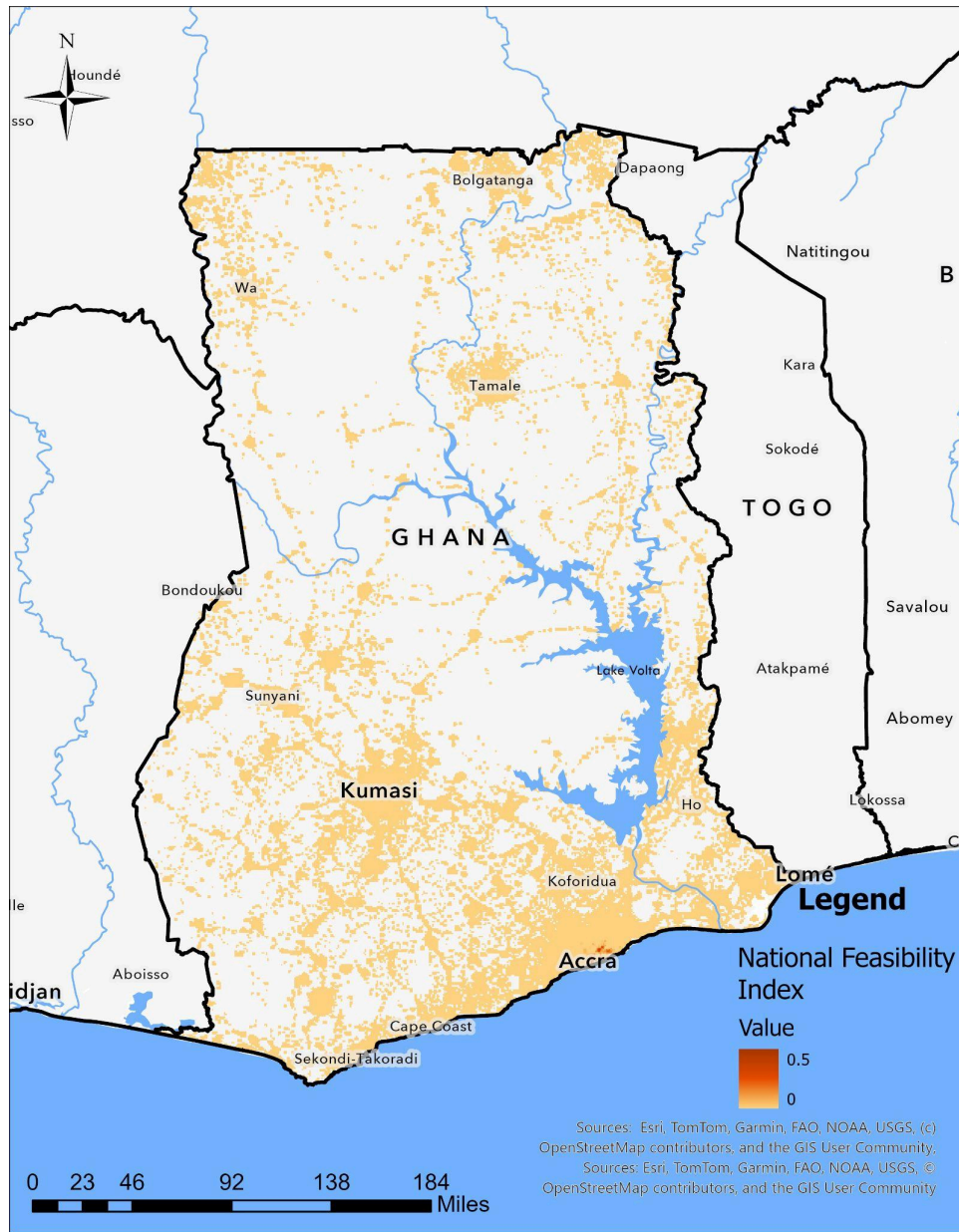


Figure: Ghana Feasibility Index

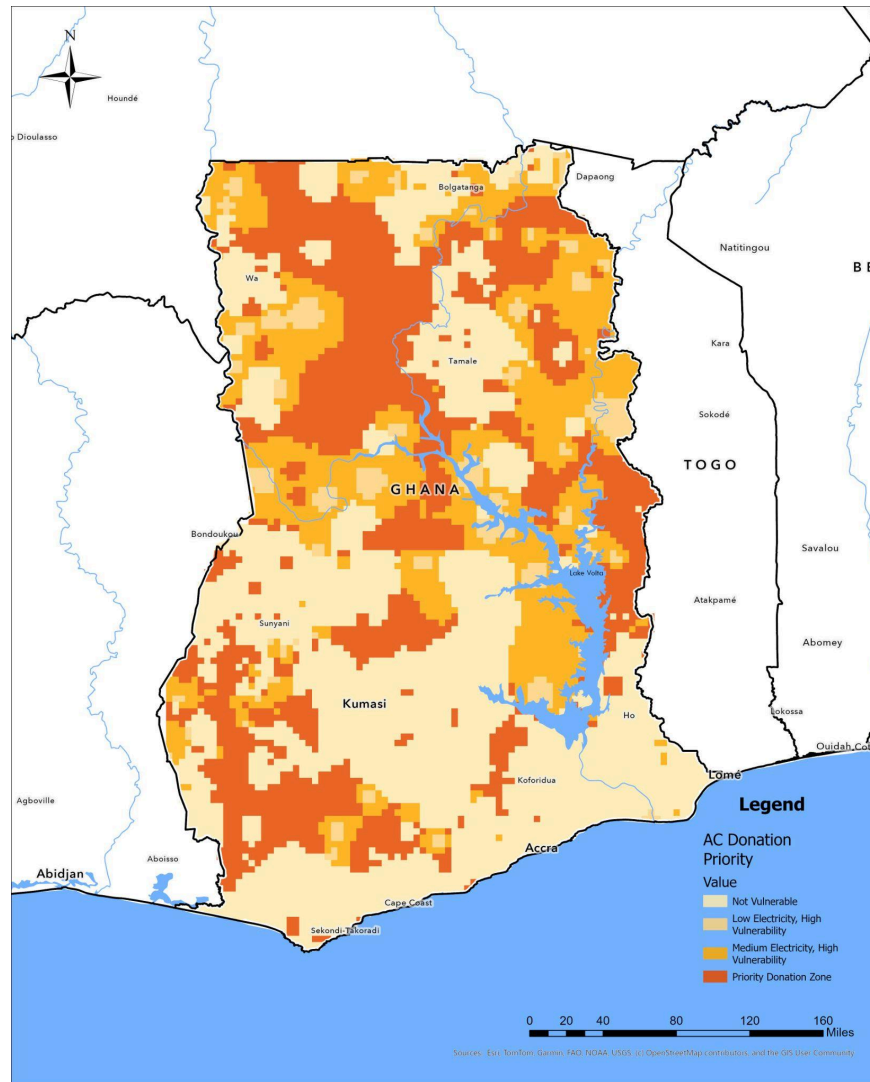


Figure: AC Donation Priority

Modelling Water Condensation of AC Units

Energy Balance Status

Air Conditioning (AC) units have four components - evaporator, compressor, condenser, and expansion valve - that have distinct functions throughout the refrigeration cycle. Each component's process can be described as follows:

- **Compressor:** Isentropic compression of warm, low pressure refrigerant into hot, high-pressure vapor.
- **Condenser:** Cools and condenses the hot, high pressure vapor into a warm liquid via a fan blowing air across a large metal finned surface area. The air is warmed in this process and released outside.

- **Expansion Valve:** Adiabatic expansion of the refrigerant from a warm liquid into a cool mist by dropping the pressure.
- **Evaporator:** Evaporates the low pressure liquid refrigerant into a gas by passing warm inside air past the evaporator coil. The now cold air is pushed out of the unit and back into the room.

The AC unit can be modeled via the vapor-compression cycle, a reliable, efficient, well-known thermodynamic cycle used for the thermal management of residential and commercial buildings. Figure 1 shows a visual of the plot, along with associated temperature-entropy and pressure-enthalpy plots. The description of the fluid, shown in varying colors, represents that state of the **refrigerant** within the AC unit.

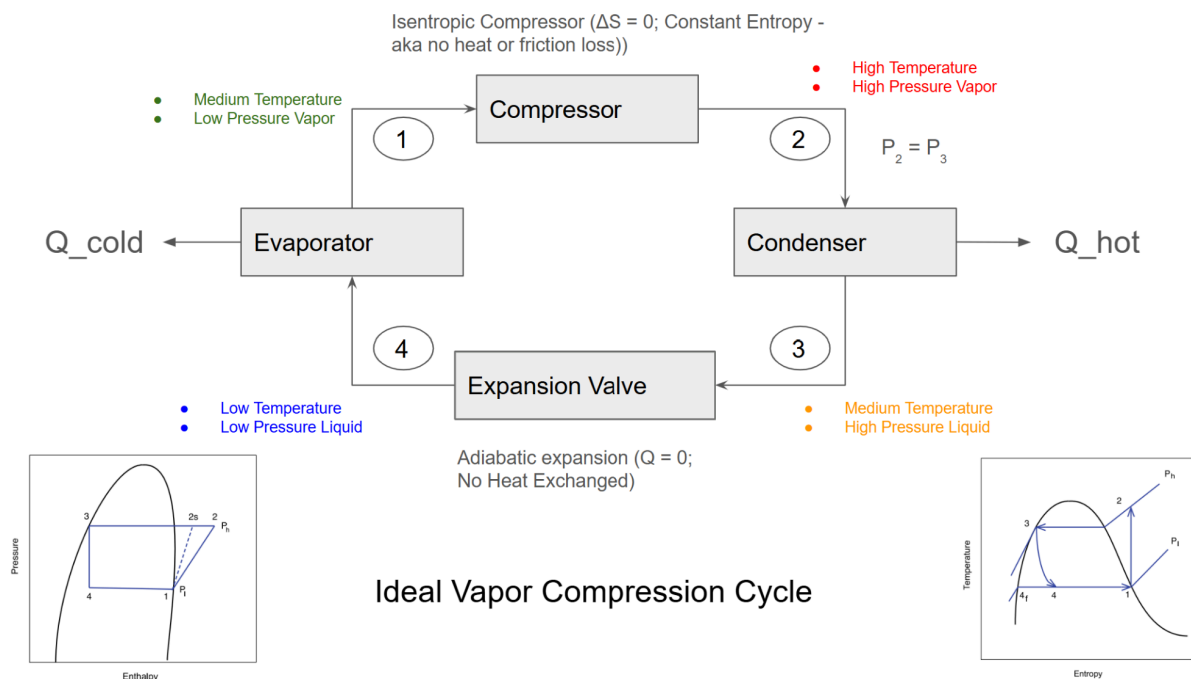


Figure 1: Preliminary Vapor - Compression Cycle demonstrating the stages of the refrigerant as it flows throughout the cycle. Corresponding pressure-enthalpy and temperature-entropy plots are from Kirkpatrick, A. (2022). *Introduction to Refrigeration and Air Conditioning Systems*.

As shown in Figure 1, the refrigerant enters the evaporator coils as a low temperature, low pressure liquid. Thus, the primary site of water condensation occurs on the evaporator coils where air is cooled below its dew point, causing the vapor to condense into droplets. Potential contaminants that may arise within this collected water include heavy metals, such as copper, lead, iron, zinc, and cadmium, along with air pollutants and bacteria that are sourced from the origin air and manifest from the moisture-rich environments of the unit itself [2]. A brief literature review is provided in the section labeled *Condensate Contamination*. Psychrometric equations can be used to evaluate the amount of water condensate that can be produced from AC units systems.

Calculating Water Condensate from Evaporator Coils

The following equation provides the rate of water condensation produced, $\dot{m}_{water}$, in the units of kilograms per second. $\dot{m}_{air}$ is the mass flow rate of air, which is also used in the following formula.

$$\dot{m}_{water} = \dot{m}_{air} (\omega_{in} - \omega_{out}) = \rho_{air} \dot{V} (\omega_{in} - \omega_{out}) \quad \text{Equation 1.1}$$

where ω_{in} and ω_{out} represent the humidity ratio at the inlet and outlet respectively. $\dot{V}$ is volumetric flow rate of air and ρ_{air} is the density of air.

The humidity ratio difference can be determined between the inlet and outlet air using psychrometric charts, which quantify the properties of air in varying environments.

Connecting Condensate Flow to Psychrometrics

Inlet Humidity

Saturation Vapor Pressure at Inlet via Tetens Equation - This represents the maximum pressure of water vapor at a given temperature in Pa. Constants are empirically derived.

$$P_{saturation\ of\ water\ @\ inlet} = 610.78 e^{(17.27 * T_{in}) / (T_{in} + 237.3)}$$

Equation 1.2

The following equation represents the actual vapor pressure at the inlet and is scaled via relative humidity. This is also in Pa.

$$P_{actual\ vapor\ @\ inlet} = RH_{inlet} * P_{saturation\ of\ water\ @\ inlet} \quad \text{Equation 1.3}$$

The following equation calculates the humidity ratio of incoming air, ω_{in} [kg/kg]. This is the unit mass of water vapor present per unit mass of dry air. The ratio of the molecular weight of water to dry air is 0.622.

$$\omega_{in} = 0.622 * (P_{actual\ vapor\ @\ inlet} / (P_{atmosphere} - P_{actual\ vapor\ @\ inlet})) \quad \text{Equation 1.4}$$

Outlet Humidity

The same procedure can be completed at the outlet as well.

Saturation Vapor Pressure at Outlet via Tetens Equation - This represents the maximum pressure of water vapor at a given temperature in Pa. Constants are empirically derived.

$$P_{saturation\ of\ water\ @\ outlet} = 610.78 e^{(17.27 * T_{out}) / (T_{out} + 237.3)} \quad \text{Equation 1.5}$$

The following equation represents the actual vapor pressure at the outlet and is scaled via relative humidity. This is also in Pa.

$$P_{actual\ vapor\ @\ outlet} = RH_{outlet} * P_{saturation\ of\ water\ @\ outlet} \quad \text{Equation 1.6}$$

The following equation calculates the humidity ratio of outgoing air, ω_{out} [kg/kg]. This is the unit mass of water vapor present per unit mass of dry air. The ratio of the molecular weight of water to dry air is 0.622.

$$\omega_{out} = 0.622 * (P_{actual\ vapor\ @\ outlet} / (P_{atmosphere} - P_{actual\ vapor\ @\ outlet})) \quad \text{Equation 1.7}$$

Bypass Factor Method

Saturation Vapor Pressure at Coil via Tetens Equation - This represents the maximum pressure of water vapor at a given temperature in Pa. Constants are empirically derived.

$$P_{saturation\ of\ water\ @\ coil} = 610.78 e^{(17.27 * T_{out}) / (T_{out} + 237.3)} \quad \text{Equation 1.8}$$

The following equation represents the actual vapor pressure at the coil, which is the same as the saturation vapor pressure. This is because RH =1 due to condensation forming. This is also in Pa.

$$P_{actual\ vapor\ @\ coil} = P_{saturation\ of\ water\ @\ coil} \quad \text{Equation 1.9}$$

The following equation calculates the humidity ratio of outgoing air, $\omega_{out, ideal}$ [kg/kg]. This represents the ideal humidity ratio leaving the unit. This is the unit mass of water vapor present per unit mass of dry air. The ratio of the molecular weight of water to dry air is 0.622.

$$\omega_{out, ideal} = 0.622 * (P_{actual\ vapor\ @\ coil} / (P_{atmosphere} - P_{actual\ vapor\ @\ coil})) \quad \text{Equation 1.10}$$

The following equation calculates the bypass factor, BF, of the AC unit. BF is a measure of the efficiency of a cooling coil within an AC unit. It represents the portion of air that passes through the coil without being cooled.

$$BF = (T_{out} - T_{coil}) / (T_{in} - T_{coil}) \quad \text{Equation 1.11}$$

The overall heat transfer coefficient, UA , of the evaporator coil can be calculated directly from the bypass factor. The relationship between the two is in the following equation. c_p is the specific heat of air.

$$BF = e^{-(UA / (\dot{m}_{air} * c_p))} \quad \text{Equation 1.12}$$

The following equation models the realistic, bypass factor-based humidity ratio, $\omega_{out, BF}$ [kg/kg]. This is the unit mass of water vapor present per unit mass of dry air. The ratio of the molecular weight of water to dry air is 0.622.

$$\omega_{out, BF} = BF * \omega_{in} + (1 - BF) (\omega_{out, ideal}) \quad \text{Equation 1.13}$$

Water Condensation Equations

Once solving for inlet humidity ratio, ω_{in} , and the two versions of outlet humidity, $\omega_{out, ideal}$ and $\omega_{out, BF}$, these values are plugged into Equation 1.1 to solve for water condensation rates for two different methods. The first method is called the “black box” method. This method uses the inlet and outlet condition and calculates the humidity ratio directly without considering internal workings of the AC unit. We believe this to be the most accurate calculation; however, this is not a very predictive model and should not be used exclusively. The second method uses the bypass factor to predict the outlet conditions. If coil conditions are known or estimated, predictions can be made to determine the water condensate amount possibility.

Total Water Condensate Amount

Calculate the total collected condensate [kg] by multiplying the condensate rate [kg/s] by the time period in seconds.

Other Variables

This model calculates the dew point, sensible/latent cooling capacities, and the sensible heat ratio, with corresponding equations detailed in the matlab code (available in github). While the project focuses on humidity ratio control under a steady air mass flow rate, future work should explore varying this rate. An optimal mass flow rate exists for each coil geometry to maximize condensate production; however, excessive rates can reduce contact time. Balancing this with thermal performance is critical, as diminished cooling efficiency is undesirable in a warming climate

Monte Carlo Simulation using Ghana's Conditions

To simulate conditions in Ghana's climate, a monte carlo model was developed to provide a probabilistic view of potential water condensation values. The following distributions were used in the model:

- Inlet temperature was modeled via normal distribution with mean 28 C and standard deviation of 4C.
- Inlet humidity was modeled via beta distribution with mean 75% and standard deviation of 15%. Beta distribution was chosen to reflect the positive, skewed behavior of relative humidity.
- Inlet Temperature was modeled via uniform distribution between 0.1 to 0.5. Uniform was used due to uncertainty of distribution of this variable. Temperature of the coil and temperature of the outlet are directly related to the bypass factor, so they follow the same distribution.

These distributions were a first attempt at the monte carlo simulation and can easily be changed. The code for this model is also provided in a github repository (linked with our project). The results are shown in the figure below. The dots represent individual runs, 1,000,000 in this run, and the color bar represents the water condensation rates possible with varying conditions. For context, with a steady rate of 0.01 kg/s of water produced for 8 hours, 288 liters (~76 gallons) of water can be potentially produced.

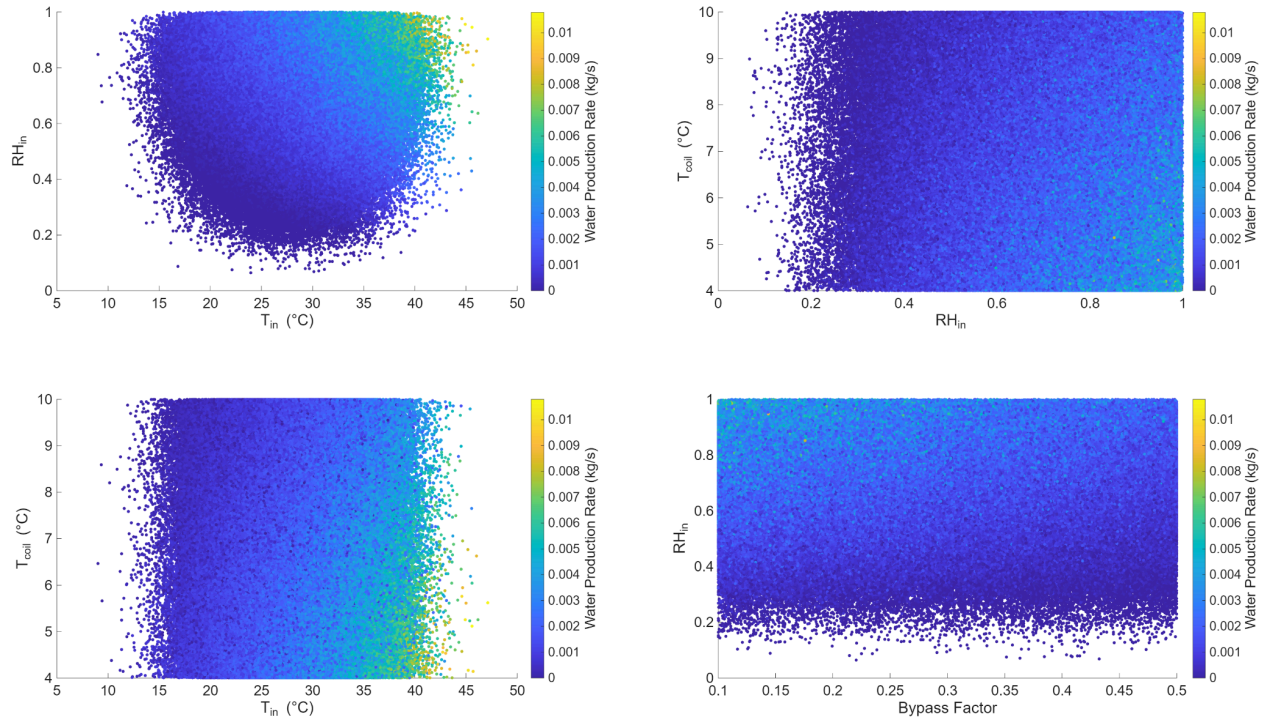


Figure: 2D Extraction Plots of the Monte Carlo Analysis to Predict Water Condensation Rates

AC Unit Selection in Ghana

Research into the types of AC units is currently being conducted within our group. In 2019, there was a study [4] completed on the various AC units within the Ghanaian AC market. Non-ducted single-split ACs are the most popular AC unit with a distinct second of

Table 3
Types and market share of air-conditioning equipment in Ghana.

Type	Percentage
Non-ducted multi-split (VRF) ACs	4%
Non-ducted single-split ACs	71%
Window-type ACs	12%
Central ACs	8%
Standing and Portable ACs	5%

window-types ACs. The figure to the right provides the exact percentages determined throughout the country, along with type. Nearly 50,000 residential air conditioners are annually sold [5] with this quantity expected to increase due to increasing global temperatures. AC units are required to have a minimum energy efficiency ratio of 2.8 to be accepted within the country's market [5]. However, many units found in Ghana were found to be unrated or 1-star in a survey completed in 2022 [6].

Due to constraints with available space and installation limitations, we opted to use a window AC unit for our testing. The window-type AC unit is the second most common type found in Ghana. Additionally, the model used works for either AC unit system type.

Materials Used and Budget

The final materials list and budget sheet can be found here: [Link](#).

Sensor Setup and Data Collection

We created a data-collection system for the temperature and humidity values at different locations on the window AC-unit. We also installed a condensation collection system to track the water generated by the window AC-unit. The collection system consists of a tube connected to the bottom of the AC unit where a hole was drilled. This tube feeds the water into a bucket that is then collected and measured after each run. The current system is built from an arduino, three SHT30 humidity and temperature sensors, a DS18B20 waterproof temperature sensor, and two thermocouples using MAX6675 amplifiers. A humidifier was placed in front of the air inlet to modify the incoming humidity levels for testing. The data is logged to an external laptop for active monitoring/graphing of the data and long term storage of the data. This collected information is then transformed into a readable format and plugged into our model for comparison. The arduino code can be found in our github repository: [AC-Hacking](#).

Sensor Placements:

1. DS18B20: in the right hand side bends of the evaporator coil.
2. Thermocouple (Front): touching the center of the front face of the evaporator coil.
3. Thermocouple (Back): touching styrofoam on the back side of the evaporator coil.
4. SHT30 Inlet: attached to the inside of the front panel.
5. SHT30 Outlet: attached to the side of the outlet, in the path of the flow.
6. SHT30 Ambient: placed atop the AC unit away from the inlet and outlet.

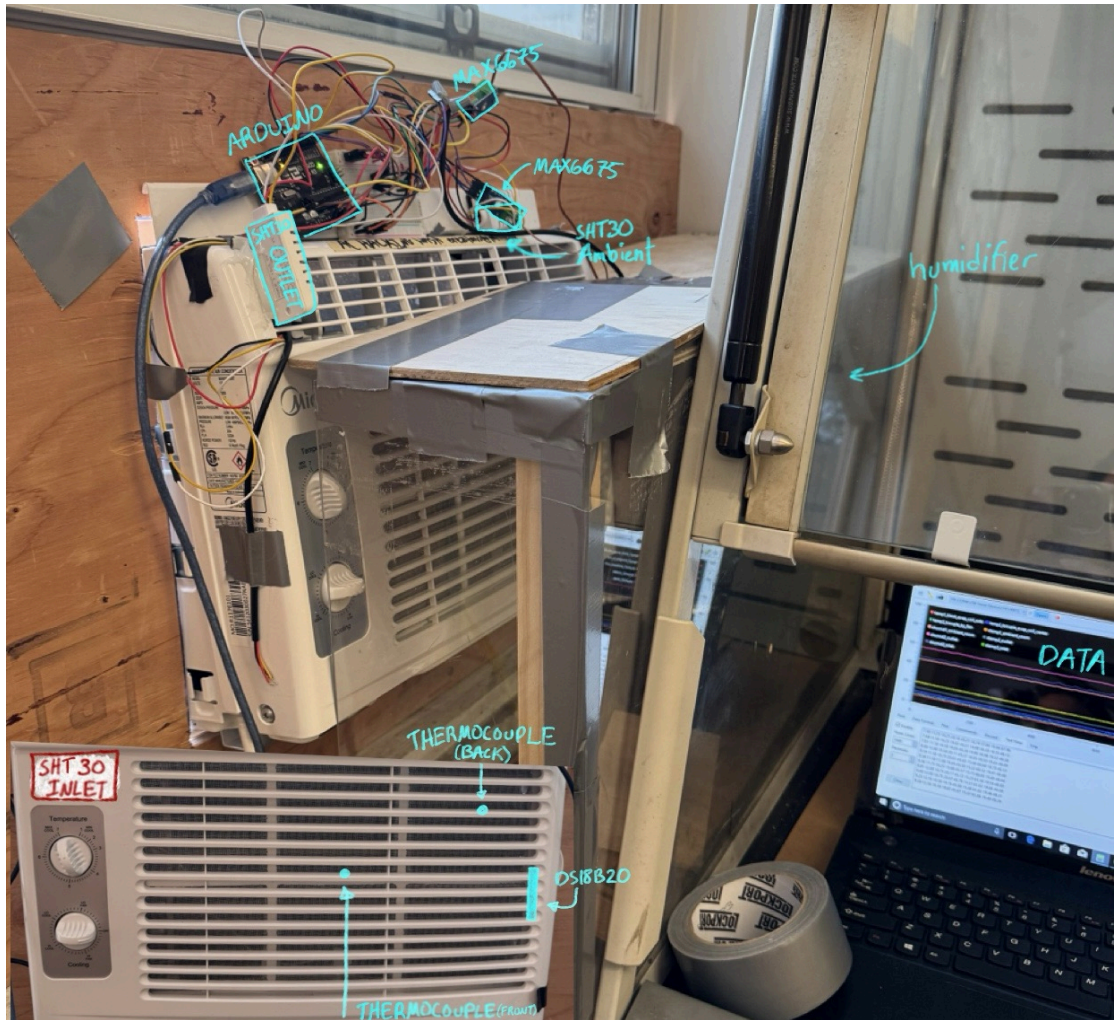


Figure: Physical setup with sensor locations.

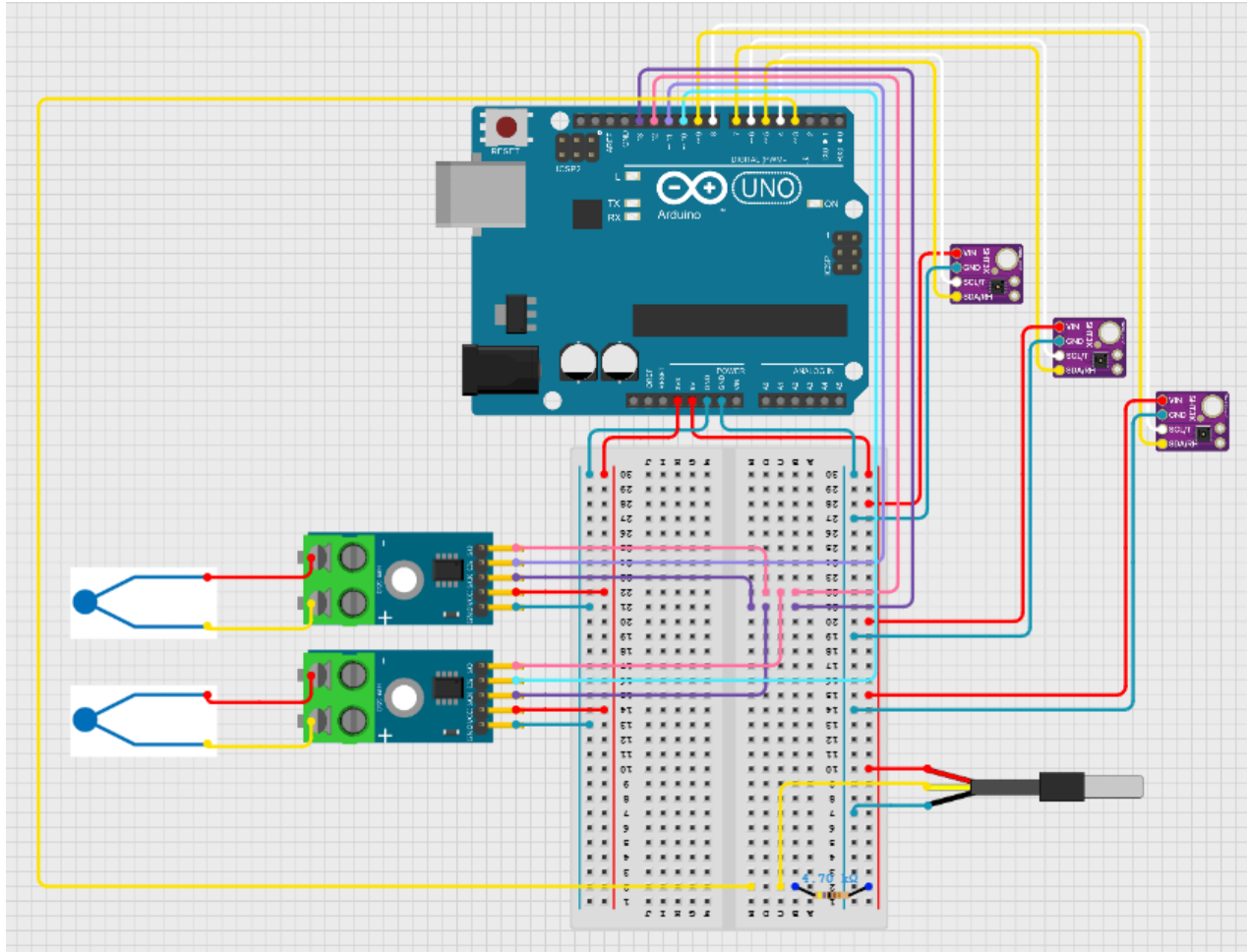


Figure: Circuit Diagram

Sensor Accuracy and Experimental Limitations

The sensors used in the experimental portion of this project come with varying temperature limits and accuracies:

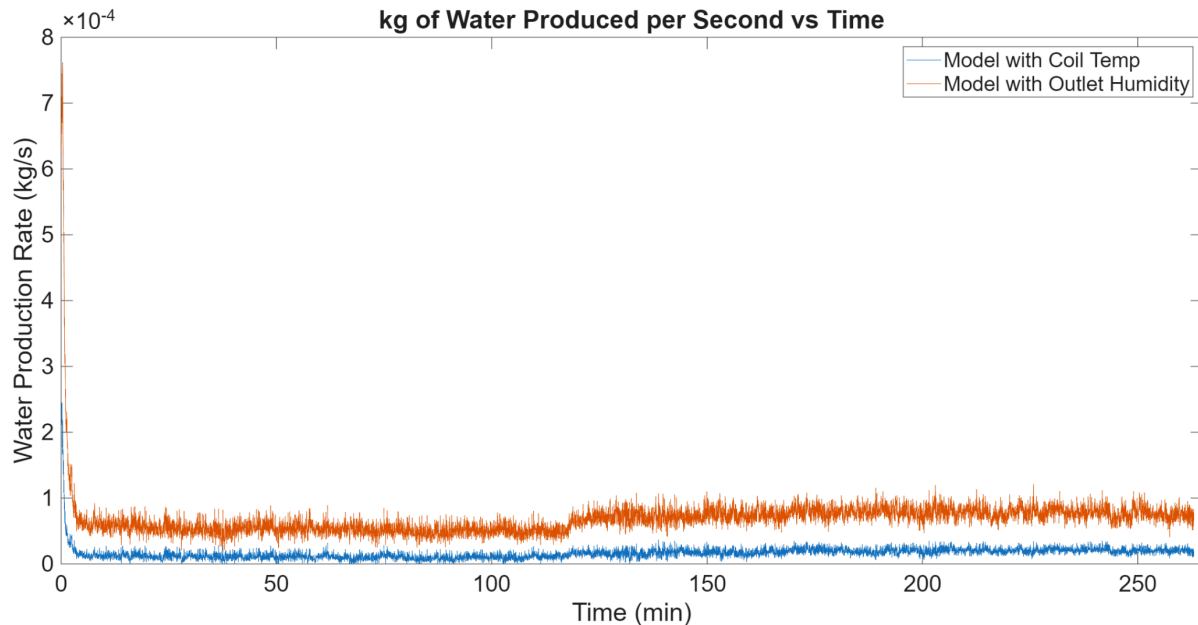
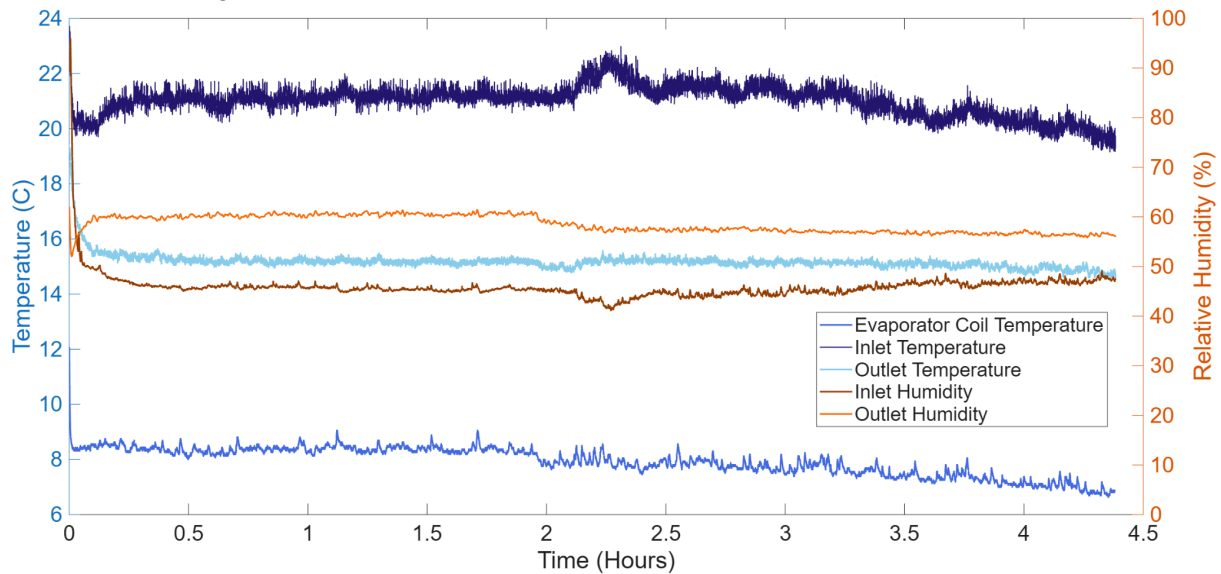
- DS18B20 temperature sensor, -10°C to $+85^{\circ}\text{C}$ with $\pm 0.5^{\circ}\text{C}$ accuracy
- Thermocouples with MAX6675, 0°C to 700°C with $\pm 1.5^{\circ}\text{C}$ accuracy
- SHT30 humidity and temperature sensor, 0% to 100% with a $\pm 2\%$ accuracy and 0°C to 65°C with a $\pm 0.2^{\circ}\text{C}$ accuracy

These accuracy ratings along with other limitations generate experimental error that must be made known:

- The hole drilled into the bottom of the AC unit collects a portion of the water, some may still go to cool the condensation coils.
- There is measurement error in water collection.
- The sensor system does not account for spatial variations.

Example Run Data and Analysis

We completed one run and the results can be shown in the figures below. Future work will need to include improved sensor placement. In the figure labelled, *Bypass Factor Vs Time*, the bypass factor of the AC unit was estimated at higher values than expected. This indicates some improper sensing locations.



Note, for the figure, *Water Produced vs Time*, the model that uses outlet humidity was far more accurate than the bypass factor for this run. Including sensor error range, listed in *Sensor Accuracy and Experimental Limitations*, the predicted value, using the “black box” model, had

less than a percent error of 20%. We hypothesize that with further runs and experimentation with sensor placement, that estimates from both models will become more accurate.

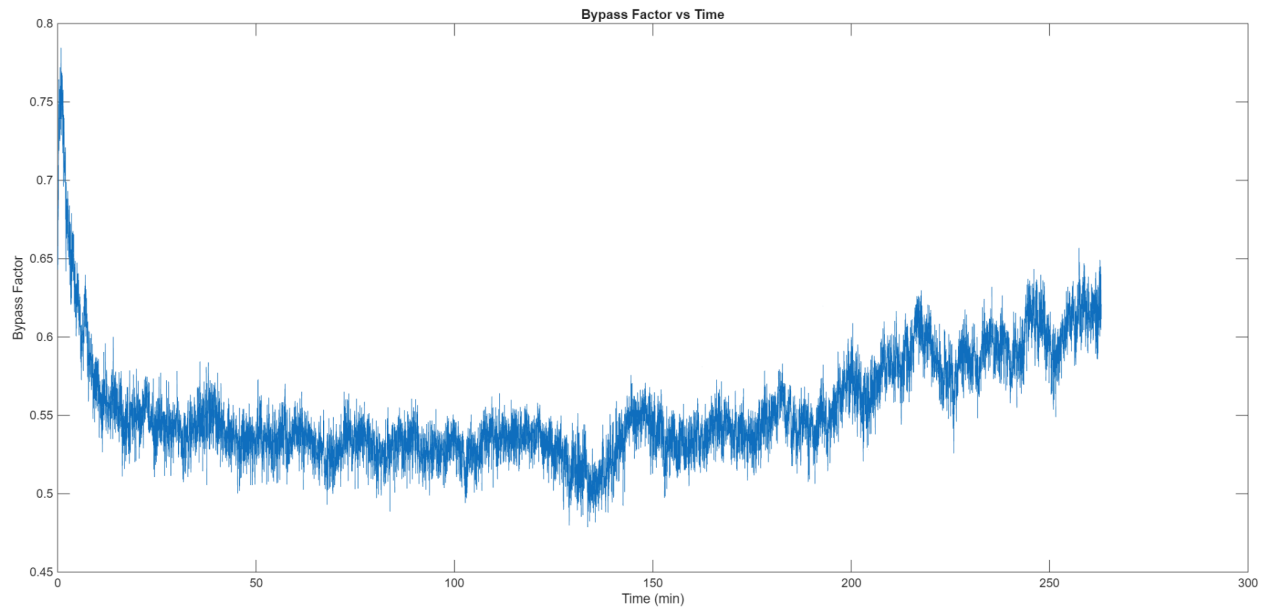


Figure: Bypass Factor vs Time

Condensate Contamination

Research results vary on if water condensate from AC units is safe for drinking. The main cause for concern in condensate water in drinking is microorganism contamination, debris collection, and above safe heavy metal concentrations. The concentration of heavy metals in the condensate water is up for debate, but researchers agree that it is safe for domestic uses beyond drinking. Future work should include sampling water for heavy metals and other contaminants to figure out exactly what the proper water treatment method is most effective or required.

Potential Contaminant Mitigation/Treatment methods

Method	Details	Notes
Regular Maintenance	Replacing air filters and cleaning coils every 2-3 months.	Proper maintenance improves indoor air quality, extends lifetime of the system, and reduces bacterial growth.
Reverse Osmosis	Forces water under pressure through a microscopic, semi-permeable membrane	RO is very effective at removing heavy metals from water. Centralized RO units

	which removes dissolved salts, heavy metals, pathogens, and many chemical contaminants.	shared across neighborhoods may reduce installation costs.
Slow Sand Filtration	Purifies water through a combination of biological, physical, and chemical processes, primarily driven by the biological layer, schmutzdecke, that forms on top of the sand.	Slow Sand Filtration kills pathogens, removes suspended solids, and reduces turbidity. One can determine the size of the filter based on the desired flow rate.
HVAC Coil Coating	Coating the evaporation coils in a thermally stable and corrosion resistant coating that will not leach contaminants into the water.	Coating the coils in can help prevent heavy metal contamination and extend the coil lifespan. They need to be properly maintained and must be thermally conductive to not prevent the cooling capability of the coils.
Ion Exchange filters	Replaces undesired ions such as heavy metals with safer ones such as sodium or hydrogen.	Ion exchange filters are particularly effective against copper, lead, and magnesium, but the filters need to be replaced periodically.
Distillation	Boiling the water and condensing the water vapor in a separate container removes most contaminants.	Heavy metals, salts, and pathogens do not carry over through distillation, but this process is energy intensive and slow. Distillation is effective on smaller scales and can be done in the homes.
Biocides	Chemical compounds that prevent biofilm buildup by killing pathogens. (ex: chlorine)	Biocides can reduce bacterial growth. When used in low concentrations, some biocides are safe to drink. It does not target heavy metals or chemical contaminants.
UV Sterilization	Exposes water to high levels of UV light to sterilize it from pathogens and prevent them from reproducing by damaging their DNA.	UV Sterilization systems are effective against pathogens, but the water should be relatively clear because suspended particles can reduce UV penetration. It does not target heavy metals or chemical contaminants.
SODIS - Solar Water Disinfection	Exposing the water to solar UV and heat kills pathogens in over the period of 6-48 hours	SODIS is simple and low-cost. It is a decentralized method for treating drinking water on the household level. It does not

		target heavy metals or chemical contaminants.
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Next Steps and Conclusions

Mapping Next Steps

The requested data from the Ghana Water Company was not received. When this GIS data arrives, Gabby would be happy to do some light continuing analysis on Accra water vulnerability conditions accordingly. Please feel free to reach out to her (gbashizi@mit.edu) to follow up!

Additionally, with the help of Dominique Bashizi, a senior software engineer, the skeleton of a Google Maps Streetview API AI application was created. The github repository of the README file he created is available [here](#). Further analysis should work on leveraging this technology in order to calculate the potential fulfillment of drinking water demand based on the number of AC units in Accra and the household need. Further understanding of the limitations of this method is required, but the established limitations are:

- Street View photos only give the longitude and latitude of the *camera*, not the object being detected
- There is some room for error in the AI detection of these units
- Not all streets in Accra have street view, in fact many in the test cells did not have photos for many streets (see [figure 3](#)).
- This analysis is very expensive and time consuming for AI to perform (figure 3 data took upwards of 45 minutes to compile)
- Strict format of long/lat must be followed in order to translate the data (AC id (yes/no) and AC count (0, 1, 2, ...)) into map form

Contacting Dominique (dom@bashizi.com) is possible to discuss and ask questions! Credit must go to Adeposi Adeogun (adeposi1@mit.edu) for the information related to his project, Waka Lagos, that leveraged this technology originally. That project is linked [here](#).

AC Unit Modelling and Sensing

Improvements in the sensing system of the AC unit may include increasing the amount of sensors and positioning the sensors in different locations. Additionally, the water collection system could include multiple holes to ensure water collection from various pathways. The modelling of the water condensate can be improved by considering the addition of transient behavior into the ac unit system.

Water Quality Testing

Water quality testing will be conducted across Accra to empirically identify contaminants in AC condensate, serving as the preliminary plan for the summer 2026 Ghana field team.

References

- [1] Kirkpatrick, Allan T. Introduction to Refrigeration and Air Conditioning Systems: Theory and Applications. Morgan & Claypool Publishers, LLC, 2022. Accessed 19 March 2026.
- [2] AlGhamdi, S.A., Alromi, D., Jambi, E. et al. Heavy metals and microbial assessment of air conditioning condensate water in Jeddah city-Saudi Arabia: concept of sustainable water resources. *Sustain. Water Resour. Manag.* 10, 139 (2024).
<https://doi.org/10.1007/s40899-024-01122-2>
- [3] “Air Conditioning - Calculating Condensate Water.” My Engineering Tools,
https://myengineeringtools.com/Air/Air_Conditioner_Condensate_Amount_Calculation_reatec_r1_1.html.
- [4] “Cooling and Heating Equations.” The Engineering ToolBox,
https://www.engineeringtoolbox.com/cooling-heating-equations-d_747.html. Accessed 19 March 2026.
- [5] Richard Opoku, Isaac Adjei Edwin, Kofi A. Agyarko, Energy efficiency and cost saving opportunities in public and commercial buildings in developing countries – The case of air-conditioners in Ghana, *Journal of Cleaner Production*, Volume 230, 2019, Pages 937-944, ISSN 0959-6526, <https://doi.org/10.1016/j.jclepro.2019.05.067>.
- [6] Samuel Gyamfi, Felix Amankwah Diawuo, Ebenezer Nyarko Kumi, Frank Sika, Mawufemo Modjinou, The energy efficiency situation in Ghana, *Renewable and Sustainable Energy Reviews*, Volume 82, Part 1, 2018, Pages 1415-1423, ISSN 1364-0321, <https://doi.org/10.1016/j.rser.2017.05.007>
- [7] Dadzie, J., Ansah, P., Apraku, K., & Pratt, I. (2022, February). Comparative analysis of air-conditioners and energy consumption levels towards green retrofitting in Ghana. In 7TH NEW ZEALAND BUILT ENVIRONMENT RESEARCH SYMPOSIUM (p. 294).
- [8] United Nations Department of Economic and Social Affairs (2025). The Sustainable Development Goals Report 2025. New York.
- [9] Ahrestani, Z., Sadeghzadeh, S., & Motejadded Emrooz, H. B. (2023). An overview of atmospheric water harvesting methods, the inevitable path of the future in water supply. *RSC Advances*, 13(15), 10273–10307. <https://doi.org/10.1039/d2ra07733g>
- [10] Zhou, X., Lu, H., Zhao, F., & Yu, G. (2020). Atmospheric Water Harvesting: A Review of Material and Structural Designs. *ACS Materials Letters*, 2(7), 671–684. <https://doi.org/10.1021/acsmaterialslett.0c00130>
- [11] Algarni, S., Saleel, C. A., & Mujeebu, M. A. (2018). Air-conditioning condensate recovery and applications—Current developments and challenges ahead. *Sustainable Cities and Society*, 37, 263–274. <https://doi.org/10.1016/j.scs.2017.11.032>
- [12] Elbrashy, A., Vafai, K., Elshennawy, A., Ayman, M., Elgebaly, A., & Rashad, M. (2025). Harvesting of Condensate Water from Air Conditioners in Large Institutions as a Sustainable

Resource. *Water Resources Management*, 39(10), 5289–5312.
<https://doi.org/10.1007/s11269-025-04203-9>

[13] Sakah, Mariette et al. (2019). A study of appliance ownership and electricity consumption determinants in urban Ghanaian households. *Sustainable Cities and Society* 44, 559-581.
<https://doi.org/10.1016/j.scs.2018.10.019>.

[14] Miguel Yanez-Barnuevo. (2025). Energy Efficiency Financing and Cooling Program in Ghana Inspires Neighbors. *Environmental and Energy Study Institute*.
<https://www.eesi.org/articles/view/energy-efficiency-financing-and-cooling-program-in-ghana-inspires-neighbors>.

[15] Acquaye, Cynthia N. A. (February 2026). *Atmospheric water & rooftop rainwater harvesting to supplement urban water supply in Ghana - The case of 3 cities* [Powerpoint slides]. Canvas@MIT

Appendix

Vapor - Compression Cycle to Water Condensate Equations

The work in this project decided to use a purely psychrometric model. However, initial brainstorming included the vapor-compression cycle as a means to connect thermal performance to water condensation amounts. We have included these equations and connections for future potential of utility.

The relevant fundamental equations [1] of an ideal vapor-compression system are as follows:

$$w_s = h_{2s} - h_1 = c_{p_{ref}} (T_{2s} - T_1) \{Isentropic\ Compression\ Work\ Per\ Unit\ Mass\} \quad \text{Equation A.0}$$

$$w_c = h_2 - h_1 = c_{p_{ref}} (T_2 - T_1) \{Actual\ Compression\ Work\ Per\ Unit\ Mass\} \quad \text{Equation A.1}$$

$$\eta_c = w_s / w_c \{Isentropic\ Efficiency\} \quad \text{Equation A.2}$$

$$q_h = h_3 - h_2 \{Heat\ Transfer\ via\ Condenser\} \quad \text{Equation A.3}$$

$$q_l = h_1 - h_4 \{Heat\ Transfer\ via\ Evaporator\} \quad \text{Equation A.4}$$

$$COP = q_l / w_c = \dot{Q}_l / \dot{W}_c \{Coefficient\ of\ Performance\} \quad \text{Equation A.5}$$

$$\dot{Q}_l = m_{refrigerant} \cdot q_l = m_{refrigerant} \cdot (h_1 - h_4) \{Cooling\ Capacity\} \quad \text{Equation A.6}$$

$$EER = 3.412 \cdot COP \{Energy\ Efficiency\ Ratio\} \quad \text{Equation A.7}$$

Considerations:

$$1) s_{2s} = s_1$$

Entropy is maintained in ideal cycle across compressor

$$2) h_3 = h_4$$

No heat transfer done in Adiabatic Expansion

Connecting Condensate Flow and Vapor Compression Equations

$$\dot{Q}_l = m_{refrigerant} \cdot q_l = m_{refrigerant} \cdot (h_1 - h_4) \quad \text{Equation A.8}$$

$$\dot{Q}_{coil} = \dot{Q}_{sensible} + \dot{Q}_{latent} = \dot{m}_{air} * c_{p_{water}} * (T_{in} - T_{out}) + \dot{m}_{water} * h_{fg} \quad \text{Equation A.9}$$

where h_{fg} is latent heat of vaporization and $c_{p_{water}}$ is heat capacity of water

$$\dot{Q}_{coil} = \dot{Q}_l$$

Thus leading to,

$$\dot{m}_{refrigerant} * (h_1 - h_4) = \dot{m}_{air} * c_{p_{water}} * (T_{in} - T_{out}) + \dot{m}_{water} * h_{fg}$$

Rearranging for mass flow rate of water (Amount of Water Collected per Unit of Time)

$$\dot{m}_{water} = \frac{\dot{m}_{refrigerant} (h_1 - h_4) - \dot{m}_{air} (c_{p_{water}})(T_{in} - T_{out})}{h_{fg}} \quad \text{Equation A.10}$$

Additional GIS Analysis and Associated Figures

Figure 1 - Temperature Map

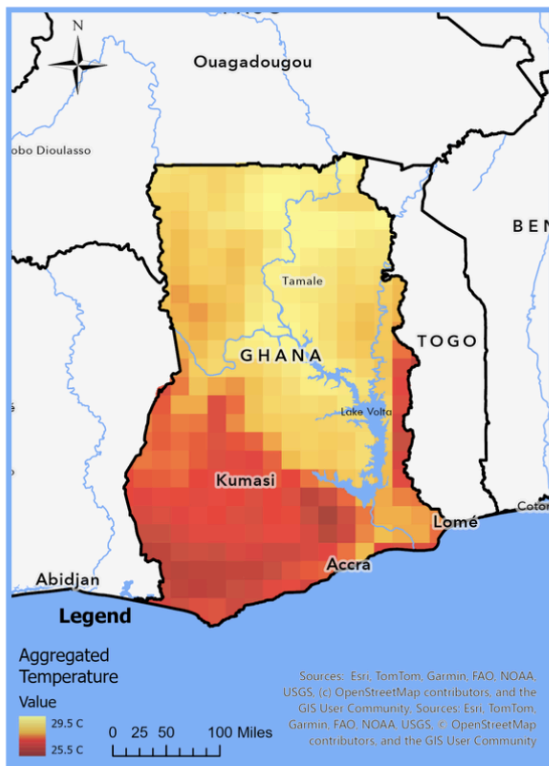


Figure 2 - Relative Humidity Map

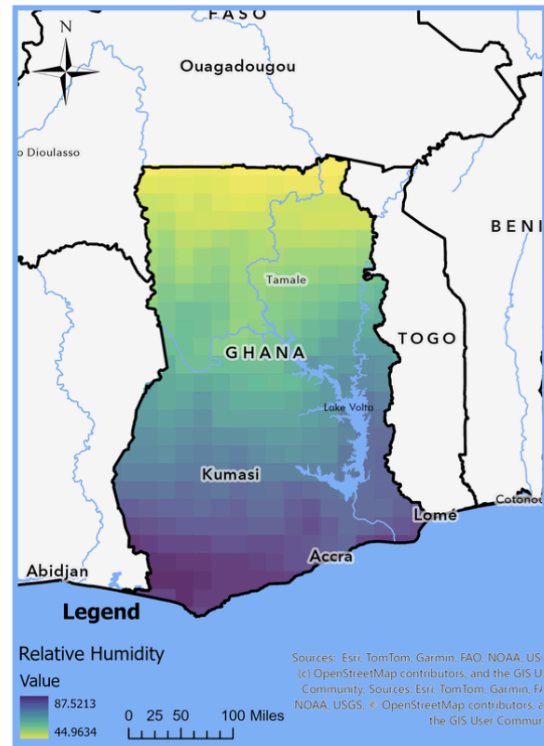


Figure 3 - Street View Test Case

